pandas concept pandas is manipulate data and analysis

```
import pandas as pd
```

```
print(pd.__version__)
```

```
2.2.2
```

```
#series
s=pd.Series([1,2,3,4,5,6])
print(s)
```

```
0    1
1    2
2    3
3    4
4    5
5    6
dtype: int64
```

```
#custom index
s=pd.Series([1,2,3,4,5,6],index=['a','b','c','d','e','f'])
print(s)
```

```
a    1
b    2
c    3
d    4
e    5
f    6
dtype: int64
```

```
# from dictonary
d={'a':1,'b':2,'c':3,'d':4,'e':5}
s=pd.Series(d)
print(s)
```

```
a    1
b    2
c    3
d    4
e    5
dtype: int64
```

```
#series operation
s=pd.Series([1,2,3,4,5,6])
print(s*2)
print(s%3)
print(s+10)
```

```
0     2
1     4
2     6
3     8
4    10
5    12
dtype: int64
0    1
1    2
2    0
3    1
4    2
5    0
dtype: int64
0    11
1    12
2    13
3    14
4    15
5    16
dtype: int64
```

```
# statistics
d=pd.Series([1,2,3,4,5,6,7,8,9,10])
print(d.max())
print(d.min())
print(d.mean())
print(d.median())
print(d.std())
print(d.describe())
```

```
10
1
5.5
5.5
3.0276503540974917
count    10.00000
mean      5.50000
std       3.02765
min       1.00000
25%       3.25000
50%       5.50000
75%       7.75000
max      10.00000
dtype: float64
```

```
d[d>5]
```

	0
5	6
6	7
7	8
8	9
9	10

dtype: int64

```
d[d>5]
```

	0
5	6
6	7
7	8
8	9
9	10

dtype: int64

```
# vectorization operation
names=pd.Series(["ibm","abu"])
print(names)
print(names.str.upper())
print(names.str.lower())
print(names.str.len())
names.str.startswith('a')
```

```
0    ibm
1    abu
dtype: object
0    IBM
1    ABU
dtype: object
0    ibm
1    abu
dtype: object
0    3
1    3
dtype: int64
```

	0
0	False
1	True

dtype: bool

```
#data frame
student={
    "name":["ibrahim","abu"],
    "age":[21,23]
}
df=pd.DataFrame(student)
print(df)
```

```
      name  age
0  ibrahim   21
1      abu   23
```

```
# list of dictonary
std=[{
    "name":"ibrahim","branch":"aiml","age":21
},
     {"name":"ibrahim","branch":"aiml"}
     ]
df=pd.DataFrame(std)
print(df)
```

```
      name branch   age
0  ibrahim   aiml  21.0
1  ibrahim   aiml   NaN
```

```
# from numpy array
import numpy as np

arr=np.random.rand(3,4)
df=pd.DataFrame(arr,columns=["a","b","c","d"],index=[1,2,3])
print(df)
```

```
          a         b         c         d
1  0.582404  0.331726  0.013724  0.939037
```

```
2  0.115766  0.316816  0.837739  0.681572
3  0.754259  0.888169  0.496294  0.413321
```

```
# data frame exploreing
data={
    "name":["ibrahim","abu","charan","vamshi"],
    "age":[21,23,25,36],
    "score":[52,36,15,24],
    "grade":["a","b","c","d"]
}
df=pd.DataFrame(data)
print(df)
```

```
      name  age  score grade
0  ibrahim   21     52     a
1      abu   23     36     b
2   charan   25     15     c
3   vamshi   36     24     d
```

```
print("head",df.head(2))
print("tail",df.tail())
print("shape",df.shape)
print("columns",df.columns)
print("index",df.index)
print("info",df.info())
print("describe",df.describe())
print("dtype",df.dtypes)
```

```
head        name  age  score grade
0  ibrahim   21     52     a
1      abu   23     36     b
tail        name  age  score grade
0  ibrahim   21     52     a
1      abu   23     36     b
2   charan   25     15     c
3   vamshi   36     24     d
shape (4, 4)
columns Index(['name', 'age', 'score', 'grade'], dtype='object')
index RangeIndex(start=0, stop=4, step=1)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   name    4 non-null      object
 1   age     4 non-null      int64 
 2   score   4 non-null      int64 
 3   grade   4 non-null      object
dtypes: int64(2), object(2)
memory usage: 260.0+ bytes
info None
describe             age      score
count   4.00000   4.000000
mean   26.25000  31.750000
std     6.70199  16.007811
min    21.00000  15.000000
```

```
25%    22.50000  21.750000
50%    24.00000  30.000000
75%    27.75000  40.000000
max    36.00000  52.000000
dtype name      object
age        int64
score      int64
grade     object
dtype: object
```

```
#selecting  columns
print(df['name'])

df.age
```

```
0    ibrahim
1        abu
2     charan
3     vamshi
Name: name, dtype: object
```

	age
0	21
1	23
2	25
3	36

dtype: int64

```
df[["name","age","score"]]
```

	name	age	score
0	ibrahim	21	52
1	abu	23	36
2	charan	25	15
3	vamshi	36	24

```
df.loc[0]
```

	0
name	ibrahim
age	21
score	52
grade	a

dtype: object

```
df.iloc[0]
```

	0
name	ibrahim
age	21
score	52
grade	a

dtype: object

```
df.loc[0:2,"name":"age"]
```

	name	age
0	ibrahim	21
1	abu	23
2	charan	25

```
df.loc[0:2]
```

	name	age	score	grade
0	ibrahim	21	52	a
1	abu	23	36	b
2	charan	25	15	c

```
df.iloc[0]
```

	0
name	ibrahim

```
df[0:2]
```

	name	age	score	grade
0	ibrahim	21	52	a
1	abu	23	36	b

```
df.iloc[-1]
```

	3
name	vamshi
age	36
score	24
grade	d

dtype: object

```
df.iloc[0:2,0:2]
```

	name	age